

GEN211 Probability and Statistics

Lecture 8: Confidence Intervals and Hypothesis Testing





Lecture contents

- Confidence intervals
- Hypothesis testing



Confidence Intervals



Sample measures (statistics) vs. population measures (parameters)

The main objective of this section is to show the procedure of estimating the value of an unknown population mean when the standard deviation of the population is known.

Suppose a college president wishes to estimate the average age of students attending classes this semester. The president could select a random sample of 100 students and find the average age of these students, say, 22.3 years. From the sample mean, the president could infer that the average age of all the students is 22.3 years. This type of estimate is called a *point estimate*.

A **point estimate** is a specific numerical value estimate of a parameter. The best point estimate of the population mean μ is the sample mean $\bar{X}$.

You might ask why other measures of central tendency, such as the median and mode, are not used to estimate the population mean. The reason is that the means of samples vary less than other statistics (such as medians and modes) when many samples are selected from the same population. Therefore, the sample mean is the best estimate of the population mean.

Sample measures (i.e., statistics) are used to estimate population measures (i.e., parameters). These statistics are called **estimators**. As previously stated, the sample mean is a better estimator of the population mean than the sample median or sample mode.



Properties of a good estimator

Three Properties of a Good Estimator

1. The estimator should be an **unbiased estimator**. That is, the expected value or the mean of the estimates obtained from samples of a given size is equal to the parameter being estimated.
2. The estimator should be consistent. For a **consistent estimator**, as sample size increases, the value of the estimator approaches the value of the parameter estimated.
3. The estimator should be a **relatively efficient estimator**. That is, of all the statistics that can be used to estimate a parameter, the relatively efficient estimator has the smallest variance.



An interval estimate

An **interval estimate** of a parameter is an interval or a range of values used to estimate the parameter. This estimate may or may not contain the value of the parameter being estimated.

In an interval estimate, the parameter is specified as being between two values. For example, an interval estimate for the average age of all students might be $21.9 < \mu < 22.7$, or 22.3 ± 0.4 years.

Either the interval contains the parameter or it does not. A degree of confidence (usually a percent) must be assigned before an interval estimate is made. For instance, you may wish to be 95% confident that the interval contains the true population mean. Another question then arises. Why 95%? Why not 99 or 99.5%?

If you desire to be more confident, such as 99 or 99.5% confident, then you must make the interval larger. For example, a 99% confidence interval for the mean age of college students might be $21.7 < \mu < 22.9$, or 22.3 ± 0.6 . Hence, a tradeoff occurs. To be more confident that the interval contains the true population mean, you must make the interval wider.



What are confidence levels and confidence intervals?

The **confidence level** of an interval estimate of a parameter is the probability that the interval estimate will contain the parameter, assuming that a large number of samples are selected and that the estimation process on the same parameter is repeated.

A **confidence interval** is a specific interval estimate of a parameter determined by using data obtained from a sample and by using the specific confidence level of the estimate.

Intervals constructed in this way are called *confidence intervals*. Three common confidence intervals are used: the 90%, the 95%, and the 99% confidence intervals.



Intuitive explanation for the confidence interval

The central limit theorem states that when the sample size is large, approximately 95% of the sample means of same-size samples taken from a population will fall within ± 1.96 standard errors of the population mean, that is,

$$\mu \pm 1.96\left(\frac{\sigma}{\sqrt{n}}\right)$$

Now, if a specific sample mean is selected, say, $\bar{X}$, there is a 95% probability that the interval $\mu \pm 1.96(\sigma/\sqrt{n})$ contains $\bar{X}$. Likewise, there is a 95% probability that the interval specified by

$$\bar{X} \pm 1.96\left(\frac{\sigma}{\sqrt{n}}\right)$$

will contain μ , as will be shown later. Stated another way,

$$\bar{X} - 1.96\left(\frac{\sigma}{\sqrt{n}}\right) < \mu < \bar{X} + 1.96\left(\frac{\sigma}{\sqrt{n}}\right)$$

Hence, you can be 95% confident that the population mean is contained within that interval when the values of the variable are normally distributed in the population.



Confidence interval formulation, revisit lecture 7, slides 19, 20

The value used for the 95% confidence interval is 1.96.

For a 99% confidence interval, the value 2.58 is used instead of 1.96 in the formula.

Since other confidence intervals are used in statistics, the symbol $z_{\alpha/2}$ (read “zee sub alpha over two”) is used in the general formula for confidence intervals. The Greek letter α (alpha) represents the total area in both tails of the standard normal distribution curve, and $\alpha/2$ represents the area in each one of the tails. The value $z_{\alpha/2}$ is called a *critical value*.

The relationship between α and the confidence level is that the stated confidence level is the percentage equivalent to the decimal value of $1 - \alpha$, and vice versa. When the 95% confidence interval is to be found, $\alpha = 0.05$, since $1 - 0.05 = 0.95$, or 95%. When $\alpha = 0.01$, then $1 - \alpha = 1 - 0.01 = 0.99$, and the 99% confidence interval is being calculated.



Confidence interval for the mean given the population standard deviation

Formula for the Confidence Interval of the Mean for a Specific α When σ is Known

$$\bar{X} - z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right) < \mu < \bar{X} + z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right)$$

For a 90% confidence interval, $z_{\alpha/2} = 1.65$; for a 95% confidence interval, $z_{\alpha/2} = 1.96$; and for a 99% confidence interval, $z_{\alpha/2} = 2.58$.

Assumptions for Finding a Confidence Interval for a Mean When σ Is Known

1. The sample is a random sample.
2. Either $n \geq 30$ or the population is normally distributed when $n < 30$.

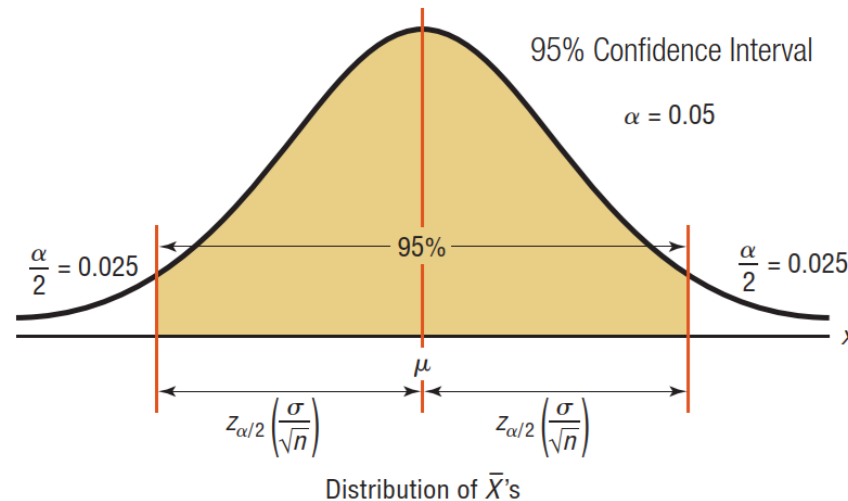
Recall lecture 7, slide 51



Confidence interval for the mean given the population standard deviation

The term $z_{\alpha/2}(\sigma/\sqrt{n})$ is called the *margin of error* (also called the *maximum error of the estimate*). For a specific value, say, $\alpha = 0.05$, 95% of the sample means will fall within this error value on either side of the population mean, as previously explained.

The **margin of error**, also called the *maximum error of the estimate*, is the maximum likely difference between the point estimate of a parameter and the actual value of the parameter.



Example 1

A researcher wishes to estimate the number of days it takes an automobile dealer to sell a Chevrolet Aveo. A random sample of 50 cars had a mean time on the dealer's lot of 54 days. Assume the population standard deviation to be 6.0 days. Find the best point estimate of the population mean and the 95% confidence interval of the population mean.

SOLUTION

The best point estimate of the population mean is 54 days. For the 95% confidence interval use $z = 1.96$.

$$\begin{aligned}\bar{X} - z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right) &< \mu < \bar{X} + z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right) \\ 54 - 1.96 \left(\frac{6.0}{\sqrt{50}} \right) &< \mu < 54 + 1.96 \left(\frac{6.0}{\sqrt{50}} \right) \\ 54 - 1.7 &< \mu < 54 + 1.7 \\ 52.3 &< \mu < 55.7 \text{ or } 54 \pm 1.7\end{aligned}$$

Hence, one can say with 95% confidence that the interval between 52.3 and 55.7 days does contain the population mean, based on a sample of 50 automobiles.



Example 2

A large department store found that it averages 362 customers per hour. Assume that the standard deviation is 29.6 and a random sample of 40 hours was used to determine the average. Find the 99% confidence interval of the population mean.

SOLUTION

The best point estimate of the population mean is 362. The 99% confidence interval for the population mean is

$$\begin{aligned}\bar{X} - z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right) &< \mu < \bar{X} + z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right) \\ 362 - 2.58 \left(\frac{29.6}{\sqrt{40}} \right) &< \mu < 362 + 2.58 \left(\frac{29.6}{\sqrt{40}} \right) \\ 362 - 12.1 &< \mu < 362 + 12.1 \\ 349.9 &< \mu < 374.1 \\ 350 &< \mu < 374\end{aligned}$$

Hence, one can be 99% confident (rounded values) that the mean number of customers that the store averages is between 350 and 374 customers per hour.



Hypothesis Testing



What is hypothesis testing?

Researchers are interested in answering many types of questions. For example, a scientist might want to know whether there is evidence of global warming. A physician might want to know whether a new medication will lower a person's blood pressure. An educator might wish to see whether a new teaching technique is better than a traditional one. A retail merchant might want to know whether the public prefers a certain color in a new line of fashion. Automobile manufacturers are interested in determining whether a new type of seat belt will reduce the severity of injuries caused by accidents. These types of questions can be addressed through statistical **hypothesis testing**, which is a decision-making process for evaluating claims about a population. In hypothesis testing, the researcher must define the population under study, state the particular hypotheses that will be investigated, give the significance level, select a sample from the population, collect the data, perform the calculations required for the statistical test, and reach a conclusion.



Definitions used in hypothesis testing

Every hypothesis-testing situation begins with the statement of a hypothesis.

A **statistical hypothesis** is a conjecture about a population parameter. This conjecture may or may not be true.

There are two types of statistical hypotheses for each situation: the null hypothesis and the alternative hypothesis.

The **null hypothesis**, symbolized by H_0 , is a statistical hypothesis that states that there is no difference between a parameter and a specific value, or that there is no difference between two parameters.

The **alternative hypothesis**, symbolized by H_1 , is a statistical hypothesis that states the existence of a difference between a parameter and a specific value, or states that there is a difference between two parameters.



Statement of the null and alternative hypotheses

As an illustration of how hypotheses should be stated, three different statistical studies will be used as examples.

Situation A A medical researcher is interested in finding out whether a new medication will have any undesirable side effects. The researcher is particularly concerned with the pulse rate of the patients who take the medication. Will the pulse rate increase, decrease, or remain unchanged after a patient takes the medication?

Since the researcher knows that the mean pulse rate for the population under study is 82 beats per minute, the hypotheses for this situation are

$$H_0: \mu = 82 \quad \text{and} \quad H_1: \mu \neq 82$$

The null hypothesis specifies that the mean will remain unchanged, and the alternative hypothesis states that it will be different. This test is called a *two-tailed test* (a term that will be formally defined later in this section), since the possible side effects of the medicine could be to raise or lower the pulse rate.



Statement of the null and alternative hypotheses

Situation B A chemist invents an additive to increase the life of an automobile battery. If the mean lifetime of the automobile battery without the additive is 36 months, then her hypotheses are

$$H_0: \mu = 36 \quad \text{and} \quad H_1: \mu > 36$$

In this situation, the chemist is interested only in increasing the lifetime of the batteries, so her alternative hypothesis is that the mean is greater than 36 months. The null hypothesis is that the mean is equal to 36 months. This test is called *right-tailed*, since the interest is in an increase only.

Situation C A contractor wishes to lower heating bills by using a special type of insulation in houses. If the average of the monthly heating bills is \$78, her hypotheses about heating costs with the use of insulation are

$$H_0: \mu = \$78 \quad \text{and} \quad H_1: \mu < \$78$$

This test is a *left-tailed test*, since the contractor is interested only in lowering heating costs.



Statement of the null and alternative hypotheses

To state hypotheses correctly, researchers must translate the *conjecture* or *claim* from words into mathematical symbols. The basic symbols used are as follows:

Equal to	=	Greater than	>
Not equal to	≠	Less than	<

The null and alternative hypotheses are stated together, and the null hypothesis contains the equals sign, as shown (where k represents a specified number).

Also notice that in a right-tailed test, the inequality sign points to the right; and in a left-tailed test, the inequality sign points to the left. Right- and left-tailed tests are also called one-tailed tests.

Two-tailed test	Right-tailed test	Left-tailed test
$H_0: \mu = k$	$H_0: \mu = k$	$H_0: \mu = k$
$H_1: \mu \neq k$	$H_1: \mu > k$	$H_1: \mu < k$



Statement of the null and alternative hypotheses

Hypothesis-Testing Common Phrases

>

Is greater than
Is above
Is higher than
Is longer than
Is bigger than
Is increased

=

Is equal to
Is the same as
Has not changed from
Is the same as

<

Is less than
Is below
Is lower than
Is shorter than
Is smaller than
Is decreased or reduced from

≠

Is not equal to
Is different from
Has changed from
Is not the same as



Example 3

State the null and alternative hypotheses for each conjecture.

- a. A researcher thinks that if expectant mothers use vitamin pills, the birth weight of the babies will increase. The average birth weight of the population is 8.6 pounds.
- b. An engineer hypothesizes that the mean number of defects can be decreased in a manufacturing process of USB drives by using robots instead of humans for certain tasks. The mean number of defective drives per 1000 is 18.
- c. A psychologist feels that playing soft music during a test will change the results of the test. The psychologist is not sure whether the grades will be higher or lower. In the past, the mean of the scores was 73.

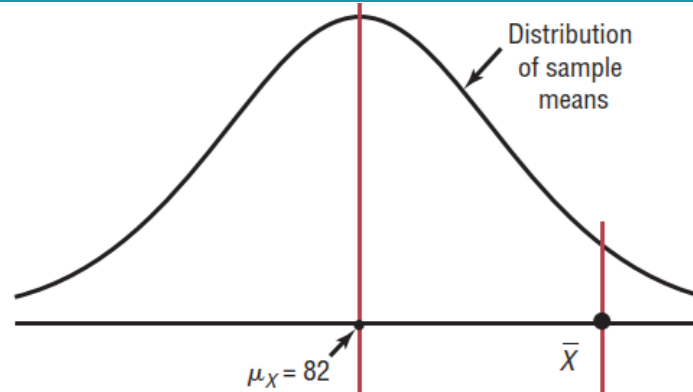
SOLUTION

- a. $H_0: \mu = 8.6$ and $H_1: \mu > 8.6$
- b. $H_0: \mu = 18$ and $H_1: \mu < 18$
- c. $H_0: \mu = 73$ and $H_1: \mu \neq 73$

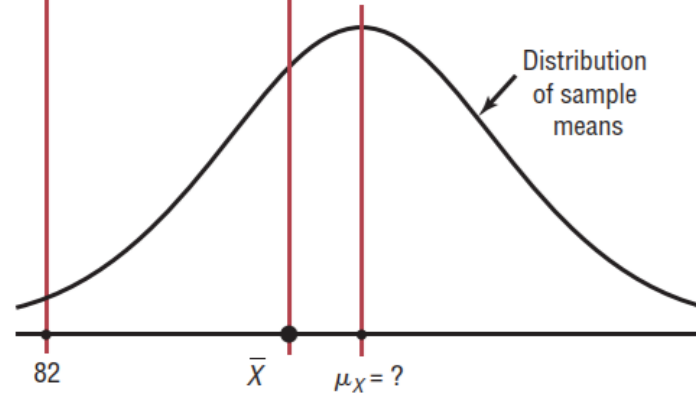


Situations in hypothesis testing, revisit slide 17

(a) H_0 is true



(b) H_0 is false



Situations in hypothesis testing, revisit lecture 7, slide 50

Recall that when samples of a specific size are selected from a population, the means of these samples will vary about the population mean, and the distribution of the sample means will be approximately normal when the sample size is 30 or more. So even if the null hypothesis is true, the mean of the pulse rates of the sample of patients will not, in most cases, be exactly equal to the population mean of 82 beats per minute. There are two possibilities. Either the null hypothesis is true, and the difference between the sample mean and the population mean is due to chance; *or* the null hypothesis is false, and the sample came from a population whose mean is not 82 beats per minute but is some other value that is not known.

The farther away the sample mean is from the population mean, the more evidence there would be for rejecting the null hypothesis. The probability that the sample came from a population whose mean is 82 decreases as the distance or absolute value of the difference between the means increases.



Statistical testing

If the mean pulse rate of the sample were, say, 83, the researcher would probably conclude that this difference was due to chance and would not reject the null hypothesis. But if the sample mean were, say, 90, then in all likelihood the researcher would conclude that the medication increased the pulse rate of the users and would reject the null hypothesis. The question is, Where does the researcher draw the line? This decision is not made on feelings or intuition; it is made statistically. That is, the difference must be significant and in all likelihood not due to chance. Here is where the concepts of statistical test and level of significance are used.

A **statistical test** uses the data obtained from a sample to make a decision about whether the null hypothesis should be rejected.

The numerical value obtained from a statistical test is called the **test value**.



Statistical testing

In this type of statistical test, the mean is computed for the data obtained from the sample and is compared with the population mean. Then a decision is made to reject or not reject the null hypothesis on the basis of the value obtained from the statistical test. If the difference is significant, the null hypothesis is rejected. If it is not, then the null hypothesis is not rejected.

As previously stated, sample data are used to determine if a null hypothesis should be rejected. Because this decision is based on sample data, there is a possibility that an incorrect decision can be made.



Possible outcomes of a hypothesis test

	H_0 true	H_0 false
Reject H_0	Error Type I	Correct decision
Do not reject H_0	Correct decision	Error Type II



Possible outcomes of a hypothesis test

In the hypothesis-testing situation, there are four possible outcomes. In reality, the null hypothesis may or may not be true, and a decision is made to reject or not reject it on the basis of the data obtained from a sample. The four possible outcomes are shown. Notice that there are two possibilities for a correct decision and two possibilities for an incorrect decision.

The four possibilities are as follows:

1. We reject the null hypothesis when it is true. This would be an incorrect decision and would result in a **type I error**.
2. We reject the null hypothesis when it is false. This would be a correct decision.
3. We do not reject the null hypothesis when it is true. This would be a correct decision.
4. We do not reject the null hypothesis when it is false. This would be an incorrect decision and would result in a **type II error**.



Error types in hypothesis testing, revisit slide 17

A **type I error** occurs if you reject the null hypothesis when it is true.

A **type II error** occurs if you do not reject the null hypothesis when it is false.

If a null hypothesis is true and it is rejected, then a *type I error* is made. In situation A, for instance, the medication might not significantly change the pulse rate of all the users in the population; but it might change the rate, by chance, of the subjects in the sample. In this case, the researcher will reject the null hypothesis when it is really true, thus committing a type I error.

On the other hand, the medication might not change the pulse rate of the subjects in the sample; but when it is given to the general population, it might cause a significant increase or decrease in the pulse rate of users. The researcher, on the basis of the data obtained from the sample, will not reject the null hypothesis, thus committing a *type II error*.



Error types in hypothesis testing, revisit slide 18

In situation B, the additive might not significantly increase the lifetimes of automobile batteries in the population, but it might increase the lifetimes of the batteries in the sample. In this case, the null hypothesis would be rejected when it was really true. This would be a type I error. On the other hand, the additive might not work on the batteries selected for the sample, but if it were to be used in the general population of batteries, it might significantly increase their lifetimes. The researcher, on the basis of information obtained from the sample, would not reject the null hypothesis, thus committing a type II error.



Hypothesis testing likened to a jury trial

The hypothesis-testing situation can be likened to a jury trial. In a jury trial, there are four possible outcomes. The defendant is either guilty or innocent, and he or she will be convicted or acquitted.

Now the hypotheses are

H_0 : The defendant is innocent.

H_1 : The defendant is not innocent (i.e., guilty).

The decision of the jury does not prove that the defendant did or did not commit the crime. The decision is based on the evidence presented. If the evidence is strong enough, the defendant will be convicted in most cases. If the evidence is weak, the defendant will be acquitted in most cases. Nothing is proved absolutely. Likewise, the decision to reject or not reject the null hypothesis does not prove anything. *The only way to prove anything statistically is to use the entire population*, which, in most cases, is not possible. The decision, then, is made on the basis of probabilities. That is, when there is a large difference between the mean obtained from the sample and the hypothesized mean, the null hypothesis is probably not true.

	H_0 true (innocent)	H_0 false (not innocent)
Reject H_0 (convict)	1. Type I error	2. Correct decision
Do not reject H_0 (acquit)	3. Correct decision	4. Type II error



How large a difference is necessary to reject the null hypothesis?

The **level of significance** is the maximum probability of committing a type I error. This probability is symbolized by α (Greek letter **alpha**). That is, $P(\text{type I error}) = \alpha$.

The probability of a type II error is symbolized by β , the Greek letter **beta**. That is, $P(\text{type II error}) = \beta$. In most hypothesis-testing situations, β cannot be easily computed; however, α and β are related in that decreasing one increases the other.

Statisticians generally agree on using three arbitrary significance levels: the 0.10, 0.05, and 0.01 levels. That is, if the null hypothesis is rejected, the probability of a type I error will be 10, 5, or 1%, depending on which level of significance is used. Here is another way of putting it: When $\alpha = 0.10$, there is a 10% chance of rejecting a true null hypothesis; when $\alpha = 0.05$, there is a 5% chance of rejecting a true null hypothesis; and when $\alpha = 0.01$, there is a 1% chance of rejecting a true null hypothesis.



Hypothesis testing terminology

The **critical** or **rejection region** is the range of test values that indicates that there is a significant difference and that the null hypothesis should be rejected.

The **noncritical** or **nonrejection region** is the range of test values that indicates that the difference was probably due to chance and that the null hypothesis should not be rejected.

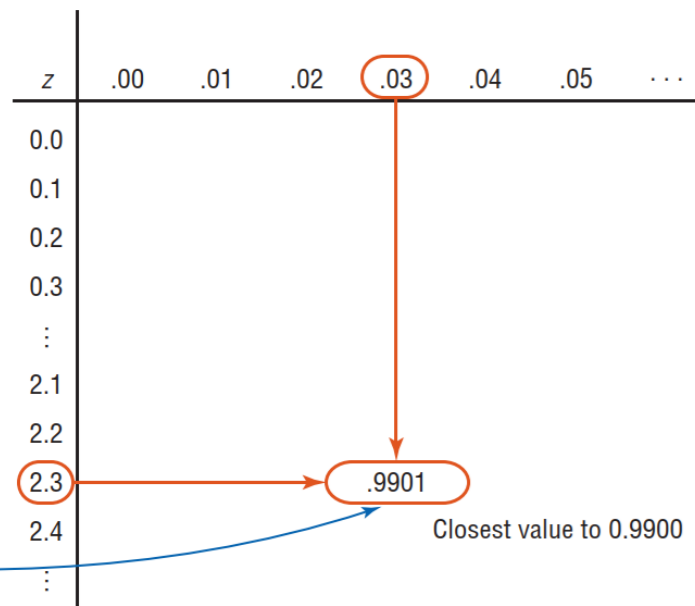
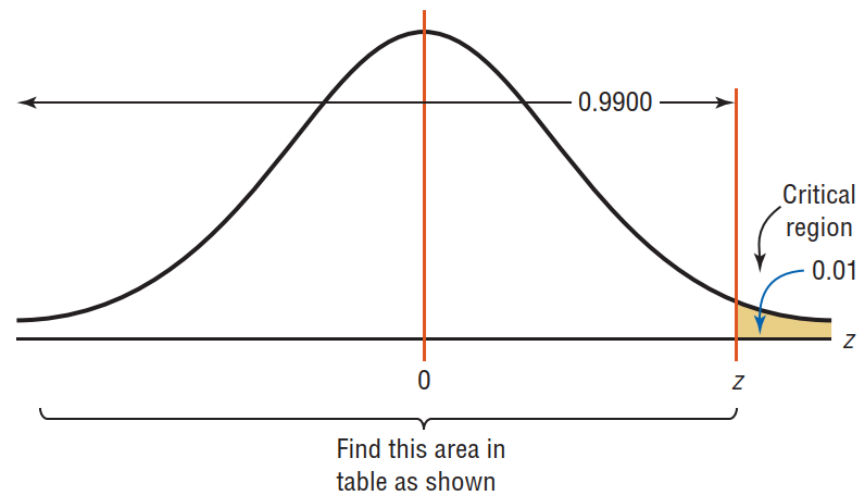
The **critical value** separates the critical region from the noncritical region. The symbol for critical value is C.V.

A **one-tailed test** indicates that the null hypothesis should be rejected when the test value is in the critical region on one side of the mean. A one-tailed test is either a **right-tailed test** or a **left-tailed test**, depending on the direction of the inequality of the alternative hypothesis.



Critical values/regions, revisit lecture 7, slide 19

Finding the Critical Value for $\alpha = 0.01$ (Right-Tailed Test)



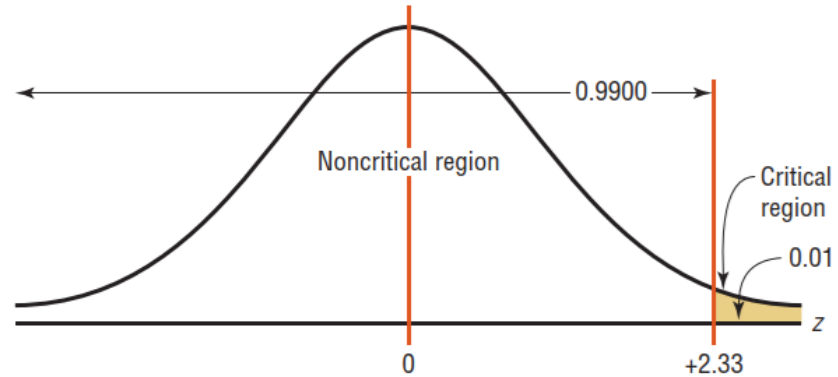
(a) The critical region

(b) The critical value

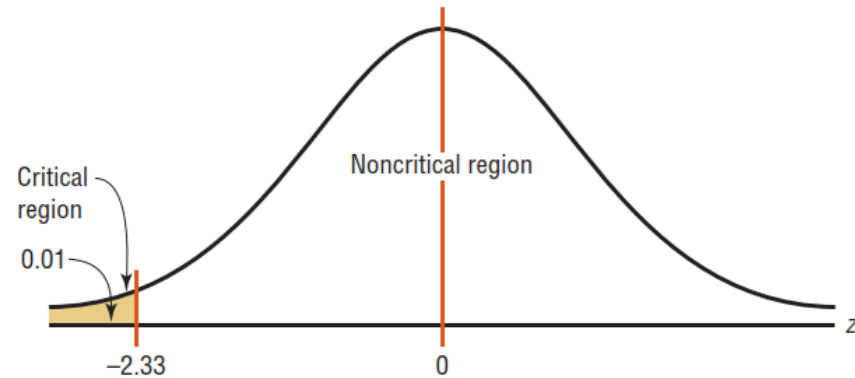


Critical values/regions, revisit lecture 7, slides 15, 19

Critical and Noncritical
Regions for $\alpha = 0.01$
(Right-Tailed Test)

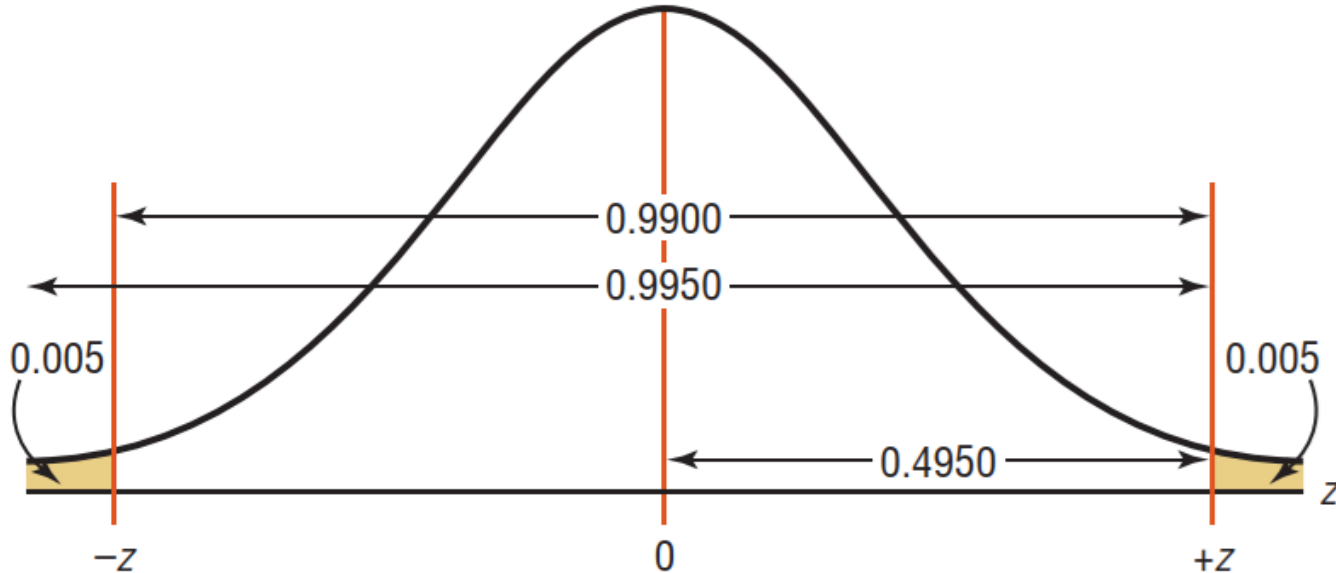


Critical and Noncritical
Regions for $\alpha = 0.01$
(Left-Tailed Test)



Critical values/regions, revisit lecture 7, slides 15, 20

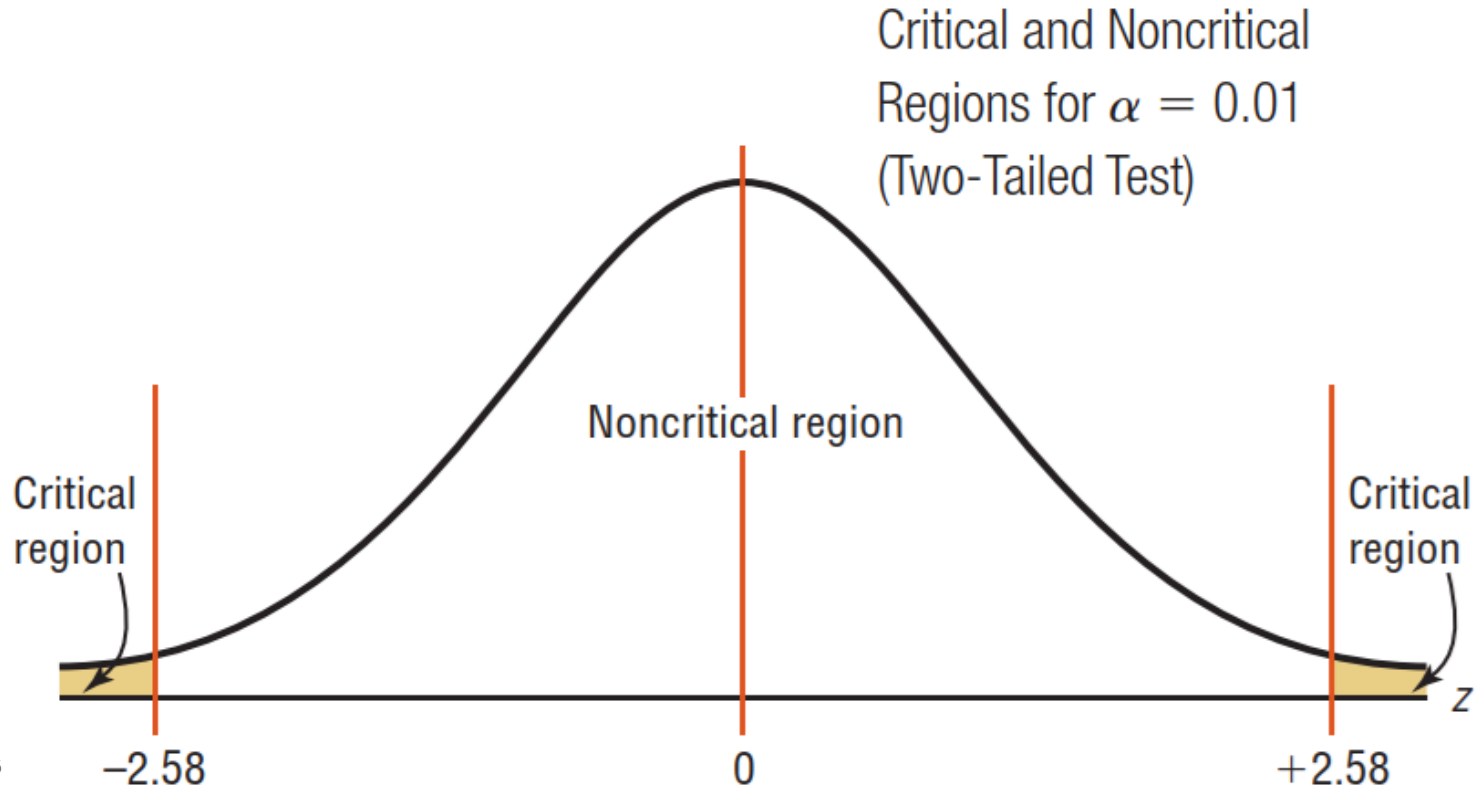
In a **two-tailed test**, the null hypothesis should be rejected when the test value is in either of the two critical regions.



Finding the Critical Values for
 $\alpha = 0.01$ (Two-Tailed Test)



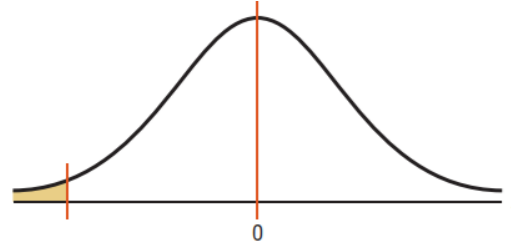
Critical values/regions, revisit lecture 7, slides 15, 20



Summary of hypothesis testing and critical values

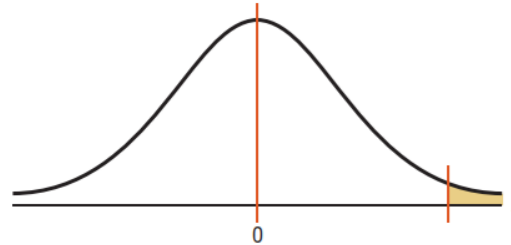
$$\begin{aligned} H_0: \mu &= k \\ H_1: \mu &< k \end{aligned} \begin{cases} \alpha = 0.10, \text{C.V.} = -1.28 \\ \alpha = 0.05, \text{C.V.} = -1.65 \\ \alpha = 0.01, \text{C.V.} = -2.33 \end{cases}$$

(a) Left-tailed



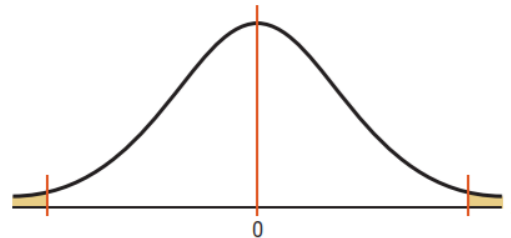
$$\begin{aligned} H_0: \mu &= k \\ H_1: \mu &> k \end{aligned} \begin{cases} \alpha = 0.10, \text{C.V.} = +1.28 \\ \alpha = 0.05, \text{C.V.} = +1.65 \\ \alpha = 0.01, \text{C.V.} = +2.33 \end{cases}$$

(b) Right-tailed



$$\begin{aligned} H_0: \mu &= k \\ H_1: \mu &\neq k \end{aligned} \begin{cases} \alpha = 0.10, \text{C.V.} = \pm 1.65 \\ \alpha = 0.05, \text{C.V.} = \pm 1.96 \\ \alpha = 0.01, \text{C.V.} = \pm 2.58 \end{cases}$$

(c) Two-tailed



Summary of hypothesis testing and critical values

Procedure Table

Finding the Critical Values for Specific α Values

- Step 1** Draw the figure and indicate the appropriate area.
- If the test is left-tailed, the critical region, with an area equal to α , will be on the left side of the mean.
 - If the test is right-tailed, the critical region, with an area equal to α , will be on the right side of the mean.
 - If the test is two-tailed, α must be divided by 2; one-half of the area will be to the right of the mean, and one-half will be to the left of the mean.
- Step 2**
- For a left-tailed test, use the z value that corresponds to the area equivalent to α .
 - For a right-tailed test, use the z value that corresponds to the area equivalent to $1 - \alpha$.
 - For a two-tailed test, use the z value that corresponds to $\alpha/2$ for the left value. It will be negative. For the right value, use the z value that corresponds to the area equivalent to $1 - \alpha/2$. It will be positive.



Summary of hypothesis testing (traditional method)

Procedure Table

Solving Hypothesis-Testing Problems (Traditional Method)

- Step 1** State the hypotheses and identify the claim.
- Step 2** Find the critical value(s).
- Step 3** Compute the test value.
- Step 4** Make the decision to reject or not reject the null hypothesis.
- Step 5** Summarize the results.

Decision	Claim	
	Claim is H_0	Claim is H_1
Reject H_0	There is enough evidence to reject the claim.	There is enough evidence to support the claim.
Do not reject H_0	There is not enough evidence to reject the claim.	There is not enough evidence to support the claim.



Example 4, revisit lecture 7, slides 15, 16, 19, 20

Find the critical value(s) for each situation and draw the appropriate figure, showing the critical region.

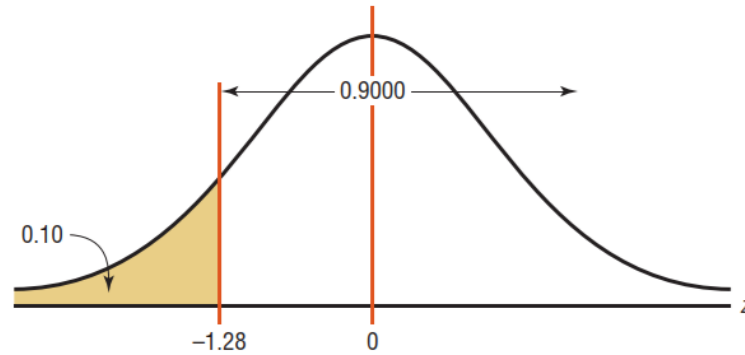
- a. A left-tailed test with $\alpha = 0.10$.
- b. A two-tailed test with $\alpha = 0.02$.
- c. A right-tailed test with $\alpha = 0.005$.

SOLUTION a

Step 1 Draw the figure and indicate the appropriate area. Since this is a left-tailed test, the area of 0.10 is located in the left tail, as shown in Figure.

Step 2 Find the area closest to 0.1000. In this case, it is 0.1003. Find the z value that corresponds to the area 0.1003. It is -1.28 .

Critical Value and Critical
Region for part a of
Example 4



Example 4

SOLUTION b

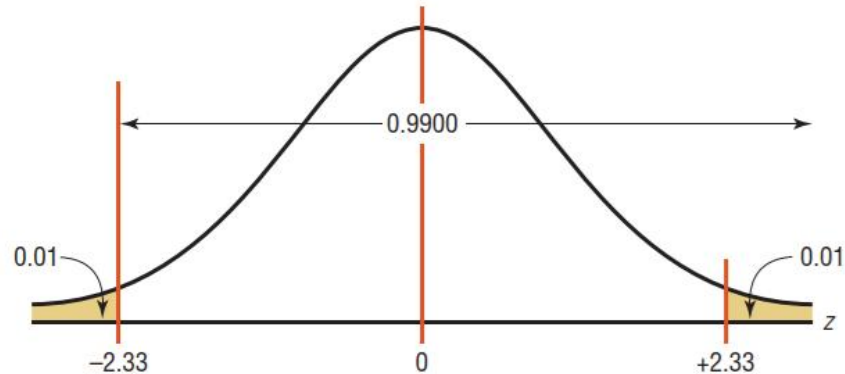
Step 1 Draw the figure and indicate the appropriate area. In this case, there are two areas equivalent to $\alpha/2$, or $0.02/2 = 0.01$.

Step 2 For the left z critical value, find the area closest to $\alpha/2$, or $0.02/2 = 0.01$. In this case, it is 0.0099.

For the right z critical value, find the area closest to $1 - \alpha/2$, or $1 - 0.02/2 = 0.9900$. In this case, it is 0.9901.

Find the z values for each of the areas. For 0.0099, $z = -2.33$. For the area of 0.9901, $z = +2.33$.

Critical Values and Critical
Regions for part b of
Example 4

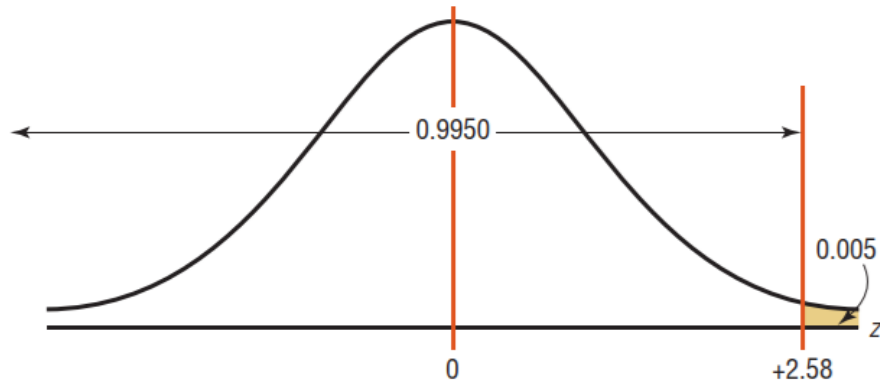


Example 4

SOLUTION c

Step 1 Draw the figure and indicate the appropriate area. Since this is a right-tailed test, the area 0.005 is located in the right tail, as shown.

Critical Value and Critical
Region for part c of
Example 4



Step 2 Find the area closest to $1 - \alpha$, or $1 - 0.005 = 0.9950$. In this case, it is 0.9949 or 0.9951.

The two z values corresponding to areas 0.9949 and 0.9951 are +2.57 and +2.58 respectively. Since 0.9950 is halfway between these two z values, find the average of the two z values $(+2.57 + 2.58) \div 2 = +2.575$. However, 2.58 is most often used.

